



MATH1123 (1004)

Calculus for Science and Engineering

Instructor: Wang Qing-Guo

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Consultation: 15:00-20:00 Monday

Teaching Assistants: Zhang Liang. Wang Yiren

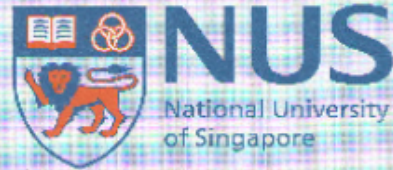
Office:

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- ✓ Studied for all degrees in Zhejiang University, China, 1978-1987
- ✓ Postdoctoral Fellow, Zhejiang University, China, 1987-1989
- ✓ Associate Professor, Zhejiang University, China, 1989-1992.
- ✓ AvH Fellow, Germany, 1990-1992
- ✓ Professor, Dept of Electrical & Computer Engineering, National University of Singapore, 1992-2015
- ✓ Distinguished Professor, University of Johannesburg, South Africa, 2015-2020
- ✓ Chair Professor, United International College, China, 2020-

My Profile

- ❖ 380 research articles in Journals
- ❖ 7 research monographs
- ❖ 20000 citations, H=80
- ❖ 5 patents
- ❖ Research funding: USD20m
- ❖ Industrial applications
- ❖ Teaching awards
- ❖ Deputy EIC for ISA Transactions (USA), on EB of five journals
- ❖ Chairman of Conferences and Societies
- ❖ Young Scientist Award, CAST, 1990
- ❖ Outstanding China-conferred Ph.D, China MoE, 1991.
- ❖ Oppenheimer grant, South Africa, 2018
- ❖ A1-rated from NRF, South Africa, from 2023
- ❖ Member of Academy of Science of South Africa, from 2017



Faculty of Engineering

Commendation List
2007/2008

is presented to

Prof Wang Qing-Guo

of the

**DEPARTMENT OF ELECTRICAL & COMPUTER
ENGINEERING**

in recognition of his dedication and commitment
to the education of the engineering students

11 November 2008

A handwritten signature in black ink, appearing to read "ESChen", is written over a horizontal line.

Dean
Faculty of Engineering



Course Objectives

- Calculus for science and engineering introduces the differential and integral calculus for univariate functions.
- It emphasizes the basic ideas and concepts on limits, derivatives, antiderivatives, definite integral, simple differential equations and corresponding applications in natural science and engineering.
- It provides the foundations for more advanced quantitative courses for science and engineering students.



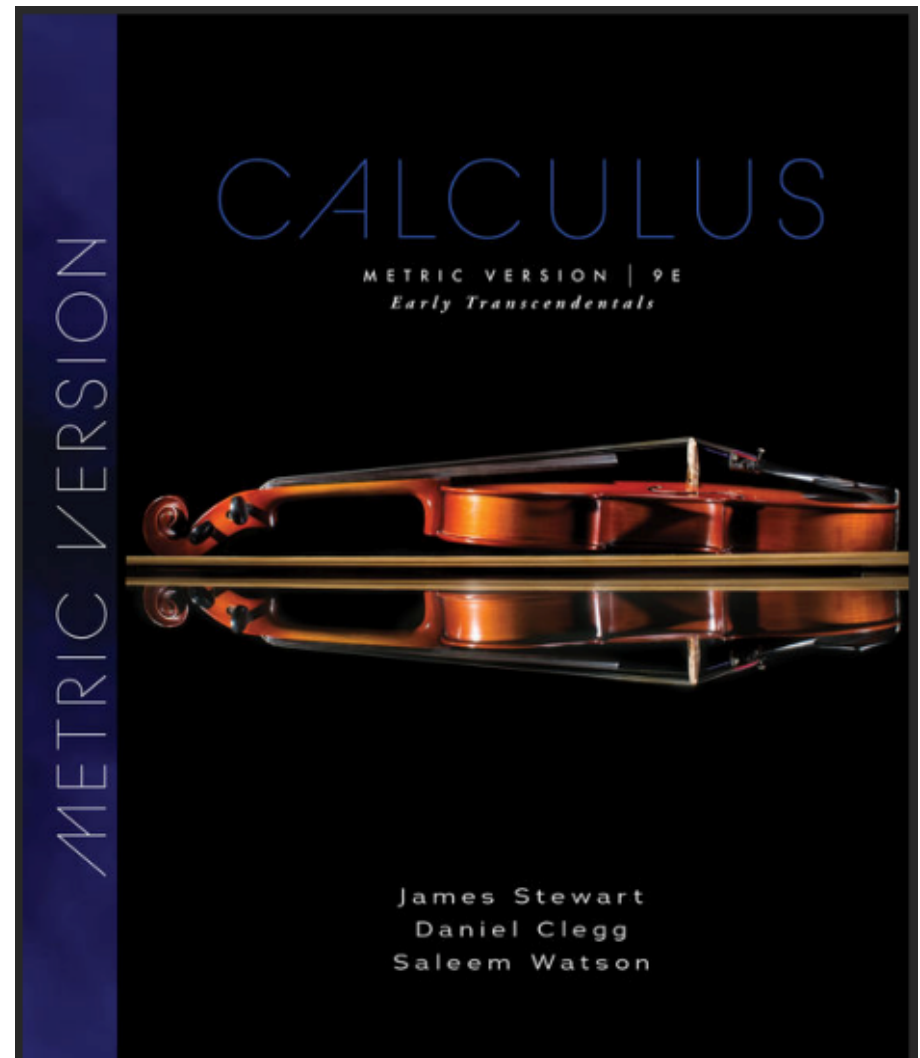
Course Schedule

1. Functions and Models (review)
2. Limits and Derivatives
3. Differential Rules
4. Application of Differentiation
5. Integrals
6. Applications of Integrations
7. Techniques of Integrations
8. Multivariable calculus (Optional)

Textbook

James Stewart, “Calculus: Early Transcendentals”, 9th Edition, Cengage Learning

For this course, we will cover Chapters 1-8 of the textbook.



References

1. S. Salas, E. Hille and G.J. Etgen, Calculus, One and Several Variables, 8th edition, John Wiley & Sons, 1999.
2. L.D. Hoffmann, G.L. Bradley, Calculus, for Business, Economics and the Social and Life sciences, 9th edition, McGraw Hill, 2004.
3. Smith and Minton, Calculus: Early Transcendental Functions. Third Edition. McGraw Hill. 2007..
4. D. Hughes-Hallett, A.M. Gleason, W.G. McCallum et al., Calculus, Single and Multivariable, 2nd edition, Wiley, 1998.
5. Ron Larson and Bruce H. Edwards, Calculus, Cengage Learning, 10th edition, 2013.
6. Michael Spivak, Calculus, Publish or Perish, 4th edition, 2008.
7. Earl W. Swokowski, Michael Olinick, Dennis Pence and Jeffery A. Cole, Calculus, Pws Pub Co, 6th edition, 1994.
8. Jon Rogawski, Calculus: Early Transcendentals, W. H. Freeman; 2nd Edition, 2011.
9. Howard Anton, Irl C. Bivens and Stephen Davis, Calculus, Wiley, 10th edition, 2012.
10. George B. Thomas Jr., Maurice D. Weir and Joel R. Hass, Thomas' Calculus, Pearson, 12th Edition, 2009.



Courseware

We use iSpace for courseware sharing



Course Conduct

Lectures: attendance and interactions

Tutorials: two subclasses conducted by TAs,
Biweekly, Time and Venue (to be confirmed)

Practice: Software simulation

Active learning, Critical thinking

Prepare for future courses



Course Assessment

Lab assignments (20%)

Written assignments: (20%)

Mid-term exam: (20%)

Final exam (40%)

Academic honesty

Copying assignments will get a **ZERO** grade